

Draft - expert review pending

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Draft translation - not yet human-reviewed / مسوده ترجمه

Computer Hardware and Software Fundamentals

A computer is two kinds of things working together: the physical parts you can touch, and the instructions that tell them what to do. This unit takes the machine apart, gently. You will learn to name the devices that put information in and get it out, the difference between the memory that forgets and the storage that remembers, the software that runs the machine and the software you run on it, and the hardware and software that let computers talk to each other across a room or a country.

Unit learning outcomes

By the end of this unit you should be able to:

- Identify **input**, **output** and **storage** devices and say what each does.
- Distinguish **primary memory** (RAM and ROM) from **secondary storage** and explain the storage hierarchy.
- Distinguish **system software**, including the operating system, from **application software** and productivity tools, and say what each is for.
- Identify the **communication hardware and software** that connect computers in a school lab or across the internet.

Prerequisite knowledge

Unit 1 (what a computer is and the classes of computers). No technical background beyond everyday use of a phone is assumed.

In this unit

1. [Topic 2.1 - Input, output, and storage devices](#) - the devices that put data in, get results out, and keep them for later.
2. [Topic 2.2 - Memory inside and outside the computer](#) - RAM, ROM and the storage hierarchy.
3. [Topic 2.3 - System software and application software](#) - what runs the machine and what you run on it.
4. [Topic 2.4 - Communication hardware and software](#) - how computers talk: networks, cables, routers and the software behind them.

How to use this unit

The four topics move from the parts you can point at (devices) to the parts you cannot (instructions and signals). Keep the glossary terms at hand; this unit introduces more new vocabulary than any other. Wherever you have access to a real computer or phone, stop and find the part being described on the actual machine - naming real parts is the fastest way to make this vocabulary stick.

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